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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 18

B.Tech.(EE)/(Electrical & Electronics Engg.) (2012 Onwards) /
(Electronics & Electrical Engg.) (2012 to 2017)
(Sem.–4)

ELECTROMAGNETIC FIELDS

Subject Code : BTEE-403
M.Code : 57106

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

Write briefly :

1) Explain the difference between the divergence and curl of a vector field.

2) State Helmholtz Theorem.

3) Define Equipotential Surfaces.

4) What is electric field intensity?

5) What is Lorentz force equation?

6) Define Magnetic Vector Potential.

7) Explain the significance of equation of continuity.

8) What is static field?

9) What is skin effect?

10) What is surface impedance?

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SECTION-B

11) State and prove the Green's Theorem.

12) Discuss Analogies between electric and magnetic fields.

13) Discuss energy stored in magnetic fields.

14) In free space, electric field density, $D = 10\sin(\omega t + \beta z\} a_x$. find magnetic flux density B, E and H.

15) State and discuss boundary conditions.

SECTION-C

16) What is meant by uniform plane wave? Derive wave equation in terms of electric field and magnetic field.

17) Discuss and explain reflection of waves by a perfect dielectric (Normal incidence).

18) State and Derive Poynting Theorem.

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